

HOMS

Mini-Assessment

Motion in One Dimension

Physical Sciences | Grade 10 | Term 3

Complete every question. Total: 30 marks.

Learner name	
Date	

CAPS-aligned draft. Educator or subject-expert approval is required before classroom use.

Instructions: Show formulae, substitutions, units and direction where required. Use the supplied graphs for graph questions.

Question 1: Concepts

1.1 Define average velocity. (2)

1.2 Give one difference between a scalar and a vector quantity. (2)

1.3 A learner says: 'An object with zero velocity must have zero acceleration.' Is this always correct? Briefly justify your answer. (2)

1.4 State what a horizontal section on a position-time graph means. (2)

Question 2: Calculations

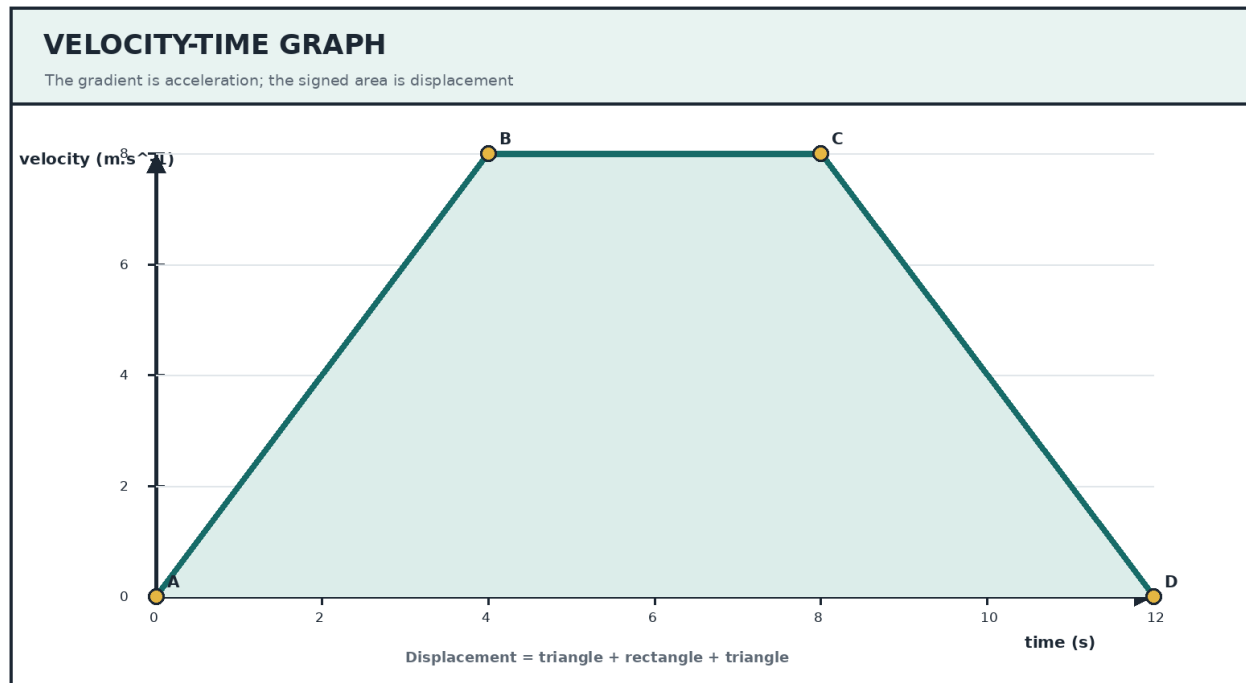
2.1 A taxi travels 120 m east and then 40 m west in 20 s. Calculate its average speed. (3)

2.2 Calculate the taxi's average velocity for the same journey. (3)

2.3 A train moving at 6 m.s^{-1} accelerates uniformly at 1.5 m.s^{-2} for 8 s. Calculate its final velocity. (2)

2.4 Calculate the train's displacement during the 8 s interval. (4)

Question 3: Graph Interpretation



3.1 Refer to the velocity-time graph. Calculate the acceleration from A to B. (2)

3.2 Describe the motion from B to C. (2)

3.3 Calculate the acceleration from C to D. (2)

3.4 Calculate the total displacement from 0 s to 12 s. (4)

TOTAL: 30